

CRITIQUE & EXPLAIN: Earthquake Magnitude, $y = 10^x$

Magnitude, x	Amplitude, y
2	100
3	1,000
?	5,500
4	10,000

Taylor	Chen
	$y = 10^x$
	$10^3 = 1,000$
	$10^4 = 10,000$
	$10^{3.5} \approx 3,162$
5,500 is halfway between 1,000 and 10,000.	$10^{3.7} \approx 5,012$
3.5 is halfway between 3 and 4.	$10^{3.8} \approx 6,310$
	$10^{3.74} \approx 5,500$
The magnitude is about 3.5.	The magnitude is about 3.74.